

Shahriar Alom Masud

Software Engineer — Backend & Applied AI

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Professional Summary

Software Engineer and author of SmartProv (v2.1.3), a published open-source library on the Arduino Registry. Experienced in developing asynchronous RESTful services and telemetry pipelines in Python, optimizing low-latency computer vision inference with ONNX Runtime, and implementing explainable AI models using Grad-CAM and TreeSHAP. Delivers end-to-end full-stack applications with relational database architecture, containerized microservices, and automated CI/CD pipelines.

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, C++, C, SQL, Bash

Backend, APIs & Databases: FastAPI, RESTful APIs, WebSockets, Supabase (PostgreSQL), Firebase, SQLite, InfluxDB Cloud, AWS

Applied AI & Computer Vision: ONNX Runtime, OpenCV, YOLOv8, MediaPipe, PyTorch, Scikit-learn, Explainable AI (Grad-CAM, TreeSHAP), Deep Learning (CNNs), Time-Series Forecasting

Frontend & Web Development: React 18/19, Tailwind CSS, Vite, Chart.js, HTML5, CSS3, Asynchronous State Management

DevOps, Tools & Automation: Docker, Git, GitHub Actions (CI/CD), Linux / Shell, Postman, n8n, PyInstaller, PyTest, System Design

Experience

Open Source Developer — SmartProv

Apr 2026 – Present

github.com/Masud744/SmartProv — C++, Asynchronous WebSockets, Flash Persistence, GitHub Actions CI/CD

- Authored and published **SmartProv (v2.1.3)** on the official package registry, providing an asynchronous, event-driven captive-portal configuration framework.
- Engineered non-volatile flash state machines with multi-network fallback and automated recovery routines, eliminating manual credential entry across 100% of device deployments.
- Adopted by developers globally; automated package distribution and semantic versioning via GitHub Actions CI/CD pipelines, reducing initial configuration time to under 30 seconds.

Technical Projects

ProctorAI — Full-Stack AI Classroom & Exam Proctoring System

Mar 2026 – Sep 2026

github.com/Masud744/ProctorAI — [Live: ai-classroom-exam-monitoring.netlify.app](https://ai-classroom-exam-monitoring.netlify.app)

Python, FastAPI, OpenCV, MediaPipe, YOLOv8, Supabase (PostgreSQL), Chart.js, PyInstaller, Netlify, Render

- Architected an automated computer vision proctoring platform running YOLOv8 and MediaPipe Face Mesh (468 landmarks) to track eye gaze (EAR), mouth aspect ratio (MAR), and 3D head pose via solvePnP at 30–60 FPS.
- Engineered a multi-threaded inference pipeline with non-blocking telemetry logging to a FastAPI and Supabase PostgreSQL backend; validated system reliability across **35+ students and 3,500+ behavioral records**.

OncoScan AI — Explainable Deep Learning Medical Diagnostic Platform

Mar 2026 – Aug 2026

github.com/Masud744/oncoscan-ai — [Live: oncoscan-ai.netlify.app](https://oncoscan-ai.netlify.app)

Python, FastAPI, PyTorch, Convolutional Neural Networks, ONNX Runtime, Grad-CAM, SQLite, Render

- Trained and deployed a 4-class brain MRI classification CNN achieving **>90% diagnostic accuracy**; integrated Grad-CAM visual heatmaps for clinical interpretability and automated PDF report generation.
- Exported and optimized the PyTorch model to ONNX format, reducing inference latency by **~4× using ONNX Runtime** within an asynchronous FastAPI microservice deployed on Render.

Solar-Aware HEMS — Full-Stack Energy Analytics & Time-Series ML

Jan 2026 – Aug 2026

github.com/Masud744/solar-aware-hems — [Live: solar-aware-hems.vercel.app](https://solar-aware-hems.vercel.app)

Python, FastAPI, Scikit-learn, TreeSHAP, React 19, Supabase, Plotly, Open-Meteo, Vercel

- Engineered time-series forecasting pipelines for solar generation and household load using quantile regression and empirical uncertainty margins ($k\sigma$) to calculate Safe Solar Surplus across 24-hour cycles.
- Constructed an asynchronous FastAPI backend integrated with Supabase and Open-Meteo; deployed an interactive React 19 and Plotly analytics dashboard on Vercel with TreeSHAP model explanations.

Education

University of Frontier Technology, Bangladesh

Aug 2023 – Jun 2027 (Expected)

Bachelor of Science (Engineering) in Internet of Things & Robotics Engineering

Gazipur, Bangladesh

Leadership & Activities

Head of Membership Development — University Robotics Club

Oct 2025 – Present

Organized technical workshops on Python programming, algorithm design, and software tools for 60+ junior engineering students; mentored members in version control and Git workflows.

RoboFusion 1.0 — Robo Soccer Segment Leader

Aug 2026

Coordinated technical arena management, tournament match scheduling, and rules enforcement for 20 participating inter-university robotics teams.